

Directions: Rewrite each of the following exponential functions in the equivalent general form $y = ab^x$, where a and b are positive constants.

1. $f(x) = 7^{x+2}$

$f(x) = 7^x \cdot 7^2$

$f(x) = 49 \cdot 7^x$

2. $g(x) = 5^{x-1}$

$g(x) = 5^x \cdot 5^{-1}$

$g(x) = \frac{1}{5} \cdot 5^x$

3. $h(x) = 2^{x+3}$

$h(x) = 2^x \cdot 2^3$

$h(x) = 8(2)^x$

4. $k(x) = 3^{x-3}$

$K(x) = 3^x \cdot 3^{-3}$

$= 3^x \cdot \frac{1}{27}$

$K(x) = \frac{1}{27} \cdot 3^x$

5. $p(x) = 2(4)^{x-1}$

$p(x) = 2 \cdot 4^x \cdot 4^{-1}$

$= 2 \cdot 4^x \cdot \frac{1}{4}$

$p(x) = \frac{1}{2}(4)^x$

6. $m(x) = 3^{2x}$

$m(x) = 9^x$

7. $r(x) = 4^{x/2}$

$r(x) = (4^{1/2})^x$

$r(x) = 2^x$

8. $n(x) = 8^{x/3}$

$n(x) = (8^{1/3})^x$

$n(x) = 2^x$

9. $s(x) = 5(2)^{3x}$

$s(x) = 5 \cdot (2^3)^x$

$s(x) = 5(8)^x$

10. Which of the following functions is an equivalent form of the function $f(x) = 4 \cdot 36^x$?

(A) $f(x) = 2 \cdot 6^{(x/2)}$

(B) $f(x) = 2 \cdot 6^{(2x)}$

(C) $f(x) = 4 \cdot 6^{(x/2)}$

(D) $f(x) = 4 \cdot 6^{(2x)}$

11. Which of the following functions is an equivalent form of the function $g(x) = 5 \cdot 3^{2x}$?

(A) $g(x) = 45^x$

(B) $g(x) = 5 \cdot 9^x$

(C) $g(x) = 25 \cdot 3^x$

(D) $g(x) = 25 \cdot 9^x$

12. The function h is given by $h(x)=9\cdot 4^{(x/2)}$. Which of the following is an equivalent form for $h(x)$?

$9\cdot(4^{1/2})^x$

$9\cdot 2^x$

(A) $h(x)=6\cdot 2^x$

(B) $h(x)=9\cdot 2^x$

(C) $h(x)=18\cdot 2^x$

(D) $h(x)=9\cdot 16^x$

13. The function k is given by $k(x)=a^2\cdot 4^x$, where a is a positive constant. Which of the following is an equivalent form for $k(x)$?

$a^2\cdot 2^{2x}$

$a^2(16^{1/2})^x$

(A) $k(x)=a\cdot 2^{(x/2)}$

(B) $k(x)=a^2\cdot 2^{(x/2)}$

(C) $k(x)=a\cdot 16^{(x/2)}$

(D) $k(x)=a^2\cdot 16^{(x/2)}$

14. Which of the following functions is an equivalent form of the function $p(x)=3^{-2x}$?

$\frac{1}{3^{2x}}$

$\left(\frac{1}{3^2}\right)^x$

(A) $p(x)=-(9)^x$

(B) $p(x)=(-9)^x$

(C) $p(x)=-\left(\frac{1}{9}\right)^x$

(D) $p(x)=\left(\frac{1}{9}\right)^x$

15. The function m is given by $m(x)=8\cdot 9^{(x/3)}$. Which of the following is an equivalent form for $m(x)$?

(A) $m(x)=2\cdot 3^x$

(B) $m(x)=2\cdot\left(\sqrt[3]{9}\right)^x$

(C) $m(x)=8\cdot 3^x$

(D) $m(x)=8\cdot\left(\sqrt[3]{9}\right)^x$